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## Abstract

Aging is a physiological process characterized by the loss of biological functions such as senescence and is associated with degenerative diseases. Reactive oxygen (ROS) and nitrogen (RNS) species accumulated over time cause an age-associated loss in bodily functions, and this forms the basis of the free radical theory of aging. Aging affects an organism at physical, physiological, cellular, and molecular levels that manifest in degenerative diseases. There exists a positive relationship between aging and nutraceuticals, and it has been of great research interest worldwide. A significant number of studies have highlighted the bioactivities of medicinal mushrooms—Cordyceps and Ganoderma that contribute to preventing aging and age-related disorders. Here, we discuss the different factors of aging and the role of nutraceuticals in combating them.

## Keywords

Aging · Free radical · Nutraceuticals · Cordyceps · Ganoderma

## 22.1 Introduction

Aging is a physiological process that every living organism experiences in their lifetime. Globally, an increase in life expectancy has been observed over a span of 50 years, and therefore, subsequent changes are witnessed in the mortality pattern in

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359

the countries of different socioeconomic development. With advancing age, deleterious effects occur on the biological systems that affect their proper functioning and cause degenerative diseases such as Alzheimer's disease, cancer, cardiovascular disease, and type 2 diabetes and cause vulnerability to death. Molecular damage like senescence, impaired intercellular contact, and loss of mitochondrial homeostasis occur that alter the structure and functions of cells and tissues (Singh et al. 2019; Partridge et al. 2018).

Furthermore, there is an associated risk factor of multimorbidity; i.e., at any particular given time, a person has more than one disorder. This multimorbidity may arise due to the interaction of different disorders, one disorder being impacted by the treatment of another one and between treatments of one disorder with another one (Beard et al. 2016).

Another important factor associated with the multidimensional dynamics between chronic diseases, physiological changes, and multimorbidity generates a complex health state that cannot be described using specific disease characteristics. These are called "geriatric syndromes." Frailty is one such syndrome that involves a progressive deterioration of physiological health with aging that results in increased susceptibility to various stressors and may ultimately have adverse outcomes (Beard et al. 2016). Frailty is described as different from normal aging, and it increases the vulnerability of older individuals to stressors. It thus results in decreased strength and endurance along with a decline in nutritional, physiological, and cognitive functions (Klein and Fain 2022).

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## 22.2 Characteristics of Aging

Aging effects can be described in the following terms (Zhang et al. 2015):

### 22.2.1 Physical Effects

The initial effect of aging on an organism can be observed as physical deterioration and increased susceptibility to age-associated diseases. There are concomitant changes that occur at cellular level such as in number and composition that occur during aging. In age-related diseases, cellular loss is observed due to depletion in stem cells, except in cancer where there is indefinite growth of cells.

#### 22.2.2 Physiological Effects

Body systems such as inflammation, nutrient sensing deregulation, and endocrine dysfunction regulate the physiological effects associated with aging. Inflammation plays an important role in clearing pathogens and healing wounded tissues during normal physiological conditions. However, chronic inflammation during aging

occurs that leads to diseases. For example, inflammatory pathway of nuclear factor (NF)- $\kappa$ B is involved in insulin resistance and neurodegenerative diseases, and therefore, inhibiting it helps in attenuating the onset of these diseases (Li et al. 2012a; Zhang et al. 2013). Similarly, deregulation of insulin/insulin-like growth factors and muscle-derived cytokines such as myokines is witnessed during aging (Demontis et al. 2014). Hence, dietary restriction and physical activity play an important role in maintaining health and longevity.

### 22.2.3 Changes at Cellular Level

In aging, several cellular malfunctions occur. For example, there is cellular senescence and an increase in reactive oxygen species occurs due to electron leakage from the mitochondrial respiratory chain which causes the oxidative stress in age-associated diseases. Also, there is an increase in endoplasmic reticulum stress that deregulates the protein folding along with defects in autophagy–lysosome and ubiquitin–proteasome systems causing proteotoxic stress.

### 22.2.4 Changes in Biomolecules

Biomolecules suffer malfunction with age due to disruption of cellular homeostasis and environmental factors. During aging, DNA damage occurs that causes mutations along with changes in the telomere length. Epigenetic modifications such as histone acetylation, histone modifications, and DNA methylation are also observed. Additionally, chaperone functions and regulation of proteolysis are affected during aging which further causes the mutations and protein folding errors.

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## 22.3 Free Radicals and Aging

Aging is associated with many deleterious effects that are linked with inflammation, DNA mutations, and increased oxidative stress. One of the most prominent theories of aging was “Free radical theory of Aging” given by Dr. Harman in 1956 (Harman 1956). According to his theory, with age, there is an increased accumulation of free radicals that cause damage to the biomolecules and pathological disorders leading to cell senescence and organismal aging.

Free radicals are highly reactive, unstable molecules with one or more unpaired electrons. These free radicals can either accept or donate an electron to other molecules and thus may act as oxidants or reductants. They can be produced endogenously in the body as a byproduct of metabolic reactions such as inflammatory responses, phagocytosis, and/or mitochondrial electron transport and can be caused due to exogenous sources such as drugs, xenobiotics, radiations, and environmental pollutants. Different types of free radicals are ROS including superoxide

radicals, hydroxyl radicals, peroxy radicals, alkoxy radicals, and RNS such as nitric oxide and nitrogen dioxide (Dhalaria et al. 2020).

These free radicals can be regulated via enzymatic or nonenzymatic antioxidants. Within the human body, enzymatic antioxidants function to maintain the balance between free radicals and antioxidants. Within the cells and tissues, enzymatic antioxidants are catalase, glutathione peroxidase, and superoxide dismutase (SOD). In addition, nutraceuticals, i.e., natural foods and their compounds, can act as antioxidants by neutralizing the ROS and RNS (Afanas'ev 2010; Dhalaria et al. 2020).

ROS signaling plays a significant role during normal physiological processes. However, during aging, genes such as Sirtuin, Klotho, FOXO3a, and p66shc interact with protein kinases such as Akt kinases and regulate ROS signaling (Afanas'ev 2010). Besides Akt kinases, other kinases also play an important role in these signaling pathways. For example, in their study on rats of age groups 6, 27 and 33 months, Wu et al. (2009) observed an increased blood glucose and superoxide along with phosphorylation of soleus p38-MAPK and ERK 1/2.

Silent information regulator Sirtuin has been reported to prevent the overproduction of ROS in conjunction with FOXO3a. Sirt1 has been shown to be considerably upregulated at low or moderate stress conditions (Alcendor et al. 2007), whereas Sirt3 and FOXO1 have been suggested to be involved in the mitochondrial signaling survival cascade.

FOXO3a is a transcription factor belonging to the O subclass of the forkhead family. It has been demonstrated to be activated due to ROS which further results in cellular senescence. Miyauchi et al. (2004) also reported that inhibition of FOXO3a and activation of Akt kinase caused senescence in human endothelial cells through a p53/p21-dependent pathway.

Similarly, Klotho protein may help in suppressing aging by inhibiting ROS formation. It has also been demonstrated to induce the nuclear translocation of FOXO3a where it binds to the manganese superoxide dismutase (MnSOD) promoter and further prevents ROS generation (Yamamoto et al. 2005).

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## 22.4 Nutraceuticals Against Aging

Due to the rise in novel infectious agents, associated treatment costs, and increased proportion of the elderly population in the world, there is accelerated interest in diet-based health management. Nutrition plays an important role in maintaining the individual's health and preventing chronic diseases.

Nutraceuticals are natural foods that provide physiological benefits in addition to being a source of nutrition through diet (Sharma and Padwad 2020). Different bioactive components of these nutraceuticals have been reported to have positive effects on the health of the elderly (Kalache et al. 2019). These bioactive components may exert an antiaging effect by their antioxidant activity and thus counteracting oxidative stress. Exogenous antioxidants may function by following mechanisms such as scavenging free radicals, decreasing ROS production, and/or repairing

oxidized membranes (Simioni et al. 2018). Since nutraceuticals can help in ameliorating excessive ROS burden, the elderly can benefit from the intake of antioxidants and can prevent oxidative stress that accompanies the aging process. Additionally, cellular senescence and immunosenescence that impair the immune system during aging may be counteracted by the immunotherapeutic potential of nutraceuticals (Sharma and Padwad 2020).

Nutritional factors may affect health by mediating epigenetic mechanisms. Epigenetic modifications, for example, DNA methylation, histone modifications, and noncoding microRNAs (miRNAs), assist the response to internal and external factors interacting with nutritional factors (Ribarič 2012).

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## 22.5 Antiaging Effects of Cordyceps

Cordyceps is a fungus belonging to the family Ascomycetes which commonly grows at an altitude of 14,000 ft. in the Indian Himalayan regions, China, Nepal, and Tibet. It is a highly valued mushroom, and different species such as *C. sinensis*, *C. militaris*, *C. taii*, and *C. sobolifera* from this genus have been studied for their therapeutic value. For many years, this fungus has been utilized as a traditional medicine for renal problems, hyperglycemia, and respiratory disorders. *C. sinensis* has been used to treat bacterial and fungal infections, cardiovascular diseases, cancers, oxidative stress, and aging (Das et al. 2021). These biological activities may be attributed to bioactive compounds existing in *C. sinensis* such as cordycepin, fatty acids, and flavonoids, nucleotide, polysaccharides, proteins, and sterols. Aging is linked to oxidative stress, Ji et al. (2009) reported in their study antiaging effects of water extract prepared from *C. sinensis*. D-galactose-induced aging caused symptoms such as reduced antioxidant enzymes, memory dysfunction, and decreased sexual function, and *C. sinensis* extract was able to improve these symptoms in a dose-dependent manner. Table 22.1 compiles the antiaging properties of different extracts or components of Cordyceps.

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## 22.6 Antiaging Effects of Ganoderma

Ganoderma, otherwise referred to as Lingzhi/Reishi, is a basidiomycete white rot fungus. It is frequently seen to grow in temperate and subtropical climatic regions including Asia, Africa, and Europe. *G. lucidum* has been traditionally used as an herbal medicine to promote health. It has been reported to have nonvolatile components such as fiber (59%), fat, (3–5%) protein (7–8%), carbohydrate (26–28%), and ash (1.8%). However, the major ingredients that contribute to its pharmacological effects are polysaccharides, peptidoglycans, and triterpenes found in mycelia, fruiting bodies, and spores.

Several studies point toward the antiaging or antiaging-related roles of the extracts and the bioactive components derived from these extracts. Sudheesh et al. (2010) reported in their study on aged mice, a decrease in ROS through the

**Table 22.1** Antiaging properties of different extracts or components of Cordyceps

| Extracts/Bioactive components                                 | Function                                              | Reference                                      |
|---------------------------------------------------------------|-------------------------------------------------------|------------------------------------------------|
| Cordycepin                                                    | Anti-inflammatory<br>Immunomodulatory<br>Antiaging    | Ashraf et al. (2020),<br>Kopalli et al. (2019) |
| Adenosine                                                     | Antioxidant<br>Anti-inflammatory                      | Jiang et al. (2011), Shin<br>et al. (2009)     |
| Exopolysaccharide fraction                                    | Antioxidant<br>Anti-inflammatory,<br>immunomodulatory | Huang et al. (2013), Li<br>et al. (2020)       |
| Acid polysaccharides from<br><i>C. sinensis</i>               | Antioxidant,<br>immunomodulatory                      | Chen et al. (2012), Li et al.<br>(2003)        |
| <i>C. militaris</i> polysaccharide                            | Antioxidant                                           | Zhang et al. (2020)                            |
| CP2-c2-S2 polysaccharide from<br><i>C. militaris</i>          | Antiaging,<br>immunomodulatory                        | Liu et al. (2016)                              |
| Fumosoroseain A from<br><i>C. fumosorosea</i>                 | Antiaging                                             | Wei et al. (2022)                              |
| Water soluble polysaccharide<br>CPS50-I from <i>C. gunnii</i> | Antiaging                                             | Zhu et al. (2011)                              |
| Water soluble polysaccharide from<br><i>C. taii</i>           | Antioxidant, antiaging                                | Xiao et al. (2012)                             |

antioxidant activity of ethanol extract of *G. lucidum*. Also, it could help in increasing the activity of important antioxidant enzymes for instance catalase, glutathione peroxidase, and MnSOD in the mitochondria of the heart, liver, and brain in aged mice. Yoon et al. (2013) also highlighted the anti-inflammatory activity of ethanol extract by inhibition of NF- $\kappa$ B and toll-like receptors in lipopolysaccharide-stimulated BV2 microglial cells. Similarly, the aqueous extract of *G. lucidum* improved stress tolerance and increased the longevity of *Caenorhabditis elegans* via the germline signaling pathway (Cuong et al. 2019), whereas polysaccharide derived from *G. lucidum* prolonged the lifespan of these nematodes by upregulating in Daf-16 expression through TIR-1 receptor and MAPK pathway (Chuang et al. 2009). Table 22.2 compiles the antiaging properties of different extracts or components of Ganoderma.

## 22.7 Conclusion

Nutraceuticals may exert antiaging effects mainly through antioxidant, free radical scavenging, immunomodulation, and anti-inflammatory activities. Aging is a complex process that has different impacts on different biological scales—physical, physiological, cellular, and molecular levels. These levels may help in identifying better strategies to counterbalance the stress and thus delay aging. Nutraceuticals could help alleviate these stressors and provide a way of healthy aging in organisms.

**Table 22.2** Antiaging properties of different extracts or components of Ganoderma

| Extracts/bioactive components                   | Function                                      | Reference                                   |
|-------------------------------------------------|-----------------------------------------------|---------------------------------------------|
| Reishi polysaccharide fraction 3 (RF3)          | Immunomodulatory effect<br>Lifespan extension | Chen et al. (2004), Chuang et al. (2009)    |
| <i>G. lucidum</i> polysaccharide I, II, III, IV | Antioxidant                                   | Shi et al. (2013)                           |
| <i>G. lucidum</i> polysaccharide B              | Immunomodulatory effect                       | Lei and Lin (1993), Mingchun et al. (2001)  |
| <i>G. lucidum</i> polysaccharide peptide        | Antioxidant, immunomodulatory effect          | Cao and Lin (2004)                          |
| Ganodermasides A, B, C, D                       | Antiaging                                     | Weng et al. (2010), Weng et al. (2011)      |
| RLZ8 recombinant <i>G. lucidum</i> protein      | Antiaging                                     | Liu et al. (2018)                           |
| Ethanol extract of <i>G. lucidum</i>            | Antioxidant, antiaging                        | Sudheesh et al. (2010), Ajith et al. (2009) |
| Water extract of <i>G. lucidum</i>              | Antiaging                                     | Cuong et al. (2019)                         |
| <i>G. atrum</i> polysaccharide PSG-1            | Antioxidant, immunomodulation, antiaging      | Li et al. (2012b), Yu et al. (2015)         |
| <i>G. capense</i> polysaccharides               | Antioxidant, antiaging                        | Huang et al. (2015), Yan et al. (2013)      |

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